Geometric Instruments

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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1. Geometry is measurement with a drawing

Astronomical geometry often begins with a ruler, a protractor and a scale. The sky gives angles; a baseline or known distance converts angles into lengths. The small-angle formula is the workhorse:

$$\theta\ (\mathrm{rad}) \approx \frac{s}{d}, \qquad s = d\theta. \tag{1}$$

For arcseconds, $s = d\,\theta''/206265$. This is valid when the angle is small and the geometry is approximately right-angled.

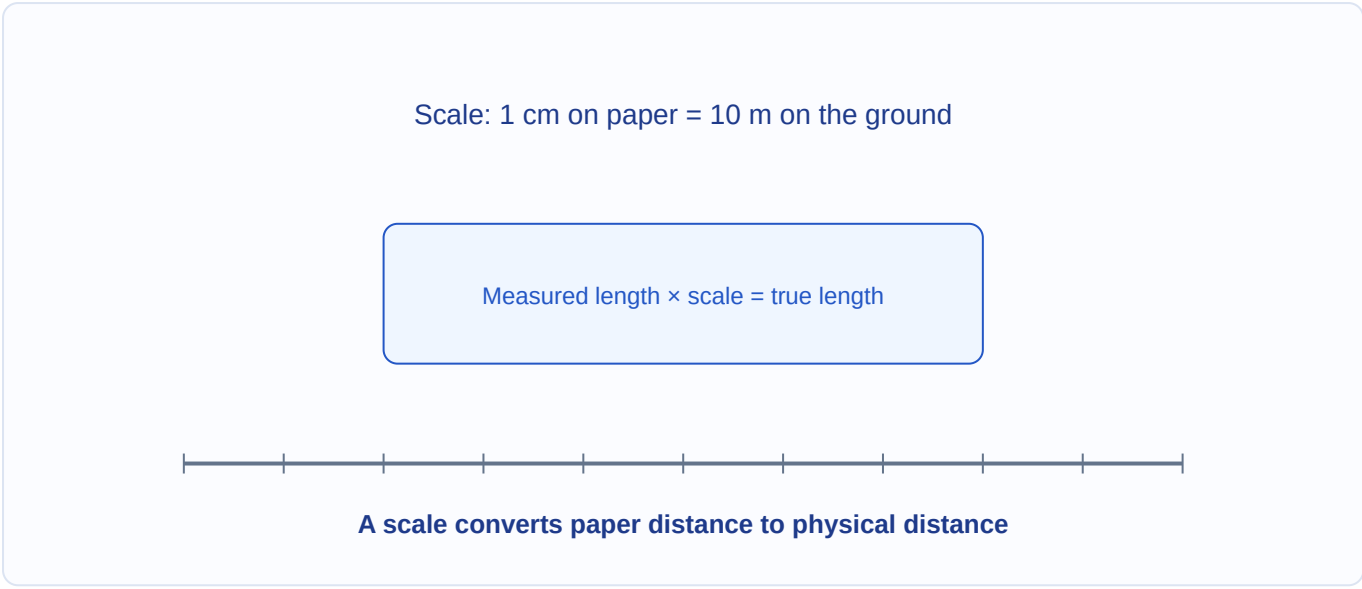


Figure 1. Always state the scale and propagate the ruler's reading uncertainty.

2. Similar triangles and scale drawings

If two triangles have the same angles, their side ratios are equal. A scale drawing preserves those ratios. If 1 cm represents 10 m, then a 3.5 cm line represents 35 m. The uncertainty in the ruler reading propagates directly through the scale.

Similar triangles explain shadow measurements, pinhole images, telescope image scale and many parallax arguments. The key is to identify which lengths correspond.

3. Trigonometry and triangulation

For a right triangle, $\sin\theta = \text{opposite}/\text{hypotenuse}$, $\cos\theta = \text{adjacent}/\text{hypotenuse}$, $\tan\theta = \text{opposite}/\text{adjacent}$. For a general triangle, the sine rule is

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}. \tag{2}$$

Triangulation uses a known baseline and two measured angles to find a distance. The same idea underlies surveying, stellar parallax and some binary-star measurements.

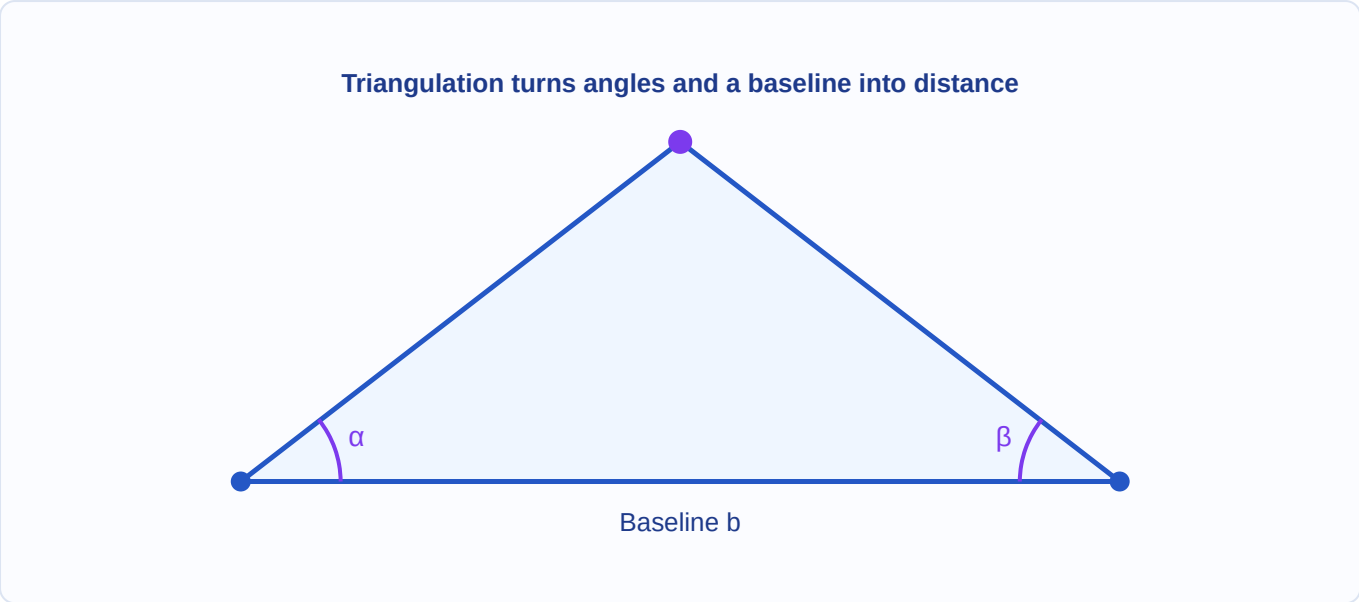


Figure 2. The same geometry underlies surveying, parallax and some astronomical distance methods.

4. Position angle and sky components

On the sky, position angle is measured from north through east. If a companion has separation ρ and position angle P, then

$$\Delta\alpha \cos\delta = \rho \sin P, \qquad \Delta\delta = \rho \cos P. \tag{3}$$

The factor $\cos\delta$ is essential: lines of right ascension converge toward the poles. A separation in RA seconds is not the same as an angular separation on the sky unless the declination is known.

5. Parallax and angular diameter

Stellar parallax uses Earth's orbital radius, 1 AU, as the baseline: the annual parallax angle p is half the peak-to-peak shift against distant background stars. If p is in arcseconds, the distance is $d = 1/p$ parsecs. Angular diameter gives physical size when distance is known: $D = d\theta$. The Moon's 0.5° diameter and 384,000 km distance imply a diameter of about 3350 km.

6. Instrumental geometry and uncertainty

A protractor has finite resolution; a ruler has a reading error; a chart has a plate scale. Convert every reading into an angular or physical uncertainty. If the plate scale is k arcsec/mm and the ruler error is σ_x mm, the angular error is $k\sigma_x$. A smaller plate scale k (fewer arcsec per mm, i.e. more mm per arcsec) reduces the angular effect of the same reading error.

INTERACTIVE EXPERIMENT

Turn a ruler-and-protractor reading into sky components.

Position angle is measured from north through east. The plate scale converts millimetres on a chart into angular separation.

Angular separation	Δα cos δ	Δδ
60.00″	**42.43″**	**42.43″**



Position angle is measured eastward from north. Δα cos δ is the east component; Δδ is the north component.

Try this. Rotate the position angle to 90°. Which component vanishes? Why is the cos δ factor needed?

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Draw the triangle. Label the known side and angles. Choose the trigonometric relation that connects them. Convert units only after the geometry is clear.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Scale drawing. A map uses 1 cm = 10 m. A line measures 3.5 cm. What physical length does it represent?

Worked solution →

[1] Problem 2 — Small angle. An object subtends 1°. Estimate its physical size at a distance of 100 m.

Worked solution →

[2] Problem 3 — Shadow height. A 1 m stick casts a 0.8 m shadow. A tree casts a 12 m shadow at the same time. Estimate the tree's height.

Worked solution →

[2] Problem 4 — Triangulation. A baseline of 100 m has angles 60° and 70° to a target. Find the distance from the 60° endpoint to the target.

Worked solution →

[1] Problem 5 — Position angle components. A companion has separation 10″ and position angle 45°. Find $\Delta\alpha \cos\delta$ and $\Delta\delta$.

Worked solution →

[2] Problem 6 — Plate scale. A chart has plate scale 2″/mm. A double-star separation measures 15 mm. Find the angular separation and the uncertainty if the ruler error is 0.5 mm.

Worked solution →

[1] Problem 7 — Parallax. A star has parallax 0.1″. Find its distance in parsecs and light-years. Use 1 pc = 3.26 ly.

Worked solution →

[2] Problem 8 — Angular diameter. The Moon's angular diameter is 0.5° and its distance is 384,000 km. Estimate its physical diameter.

Worked solution →

[3] Problem 9 — RA conversion. Two stars differ by 1 second of RA at declination 60°. What is their east–west angular separation on the sky?

Worked solution →

[3] Problem 10 — Design a measurement. You want to measure the height of a building using only a protractor, a tape measure and level ground. Describe the measurement and one way to reduce uncertainty.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.